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Learning mobility models from traffic data

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1 Introduction

1.1 Urban context and motivation

Introduce the problem of traffic disruptions (planned works, incidents, etc.) and explain why selecting detours is difficult due to network topology and traffic conditions. It motivates the need for a decision-support approach that can compare multiple recovery plans in a realistic way rather than relying on intuitions and simple metrics.

1.1.1 Digital Twin

A digital twin is a data-driven virtual representation of real-world entities designed to mirror a physical object or system throughout its lifecycle. The virtual part is updated with real-time data gotten from sensors or other sources making it possible to link the physical and digital parts. [1], [2]

A digital twin consists of three components [3]:

1. **Physical entity:** The real-world object or system being modeled (e.g., traffic, a plant)
2. **Virtual representation:** A model that replicates the behavior of the physical entity.
3. **Connection:** The link between the physical system and the virtual system making it possible to update the physical system by exchanging information via this real-time connection.

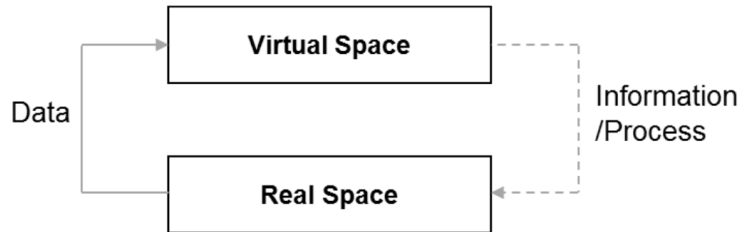


Figure 1: Illustration of a digital twin from [4]

Digital twins receive continuously streams of data from sensors or other sources ensuring that the virtual model replicates the current state of the physical object. The main goals for a digital twin are to simulate the physical entity and predict what could happen in order to optimize or predict failures. Urban traffic networks, which are both data-rich and fast-evolving, provide fertile ground for the deployment of digital twins (see **Figure 2**). Digital twins in traffic can potentially simulate and optimize traffic flows, supporting smarter city planning and real-time operations. A detailed discussion of modern digital twin solutions in traffic management is provided in **subsubsection 2.1.1**.

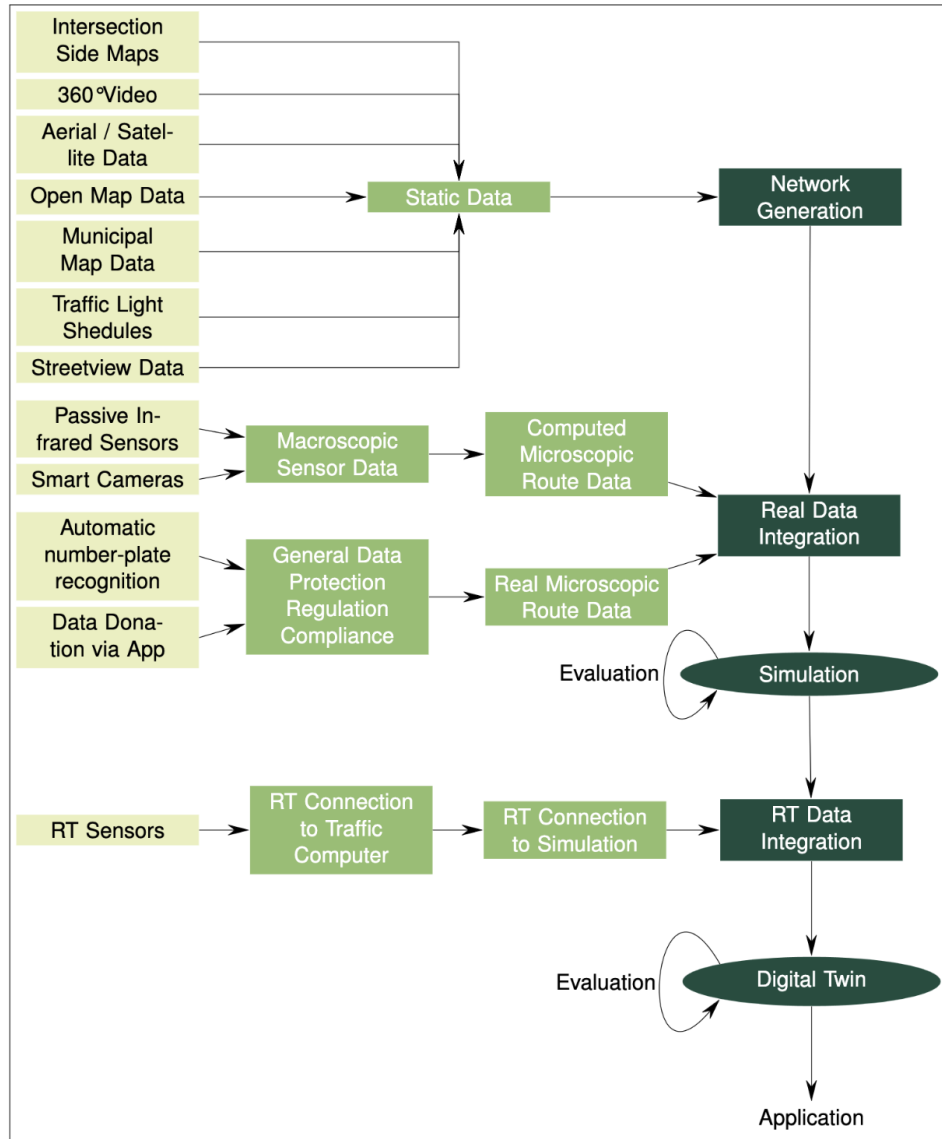


Figure 2: xample workflow for driving SUMO towards a digital twin. Various static and real-time data sources (maps, sensors, cameras) are integrated into simulation models to continuously update and calibrate the digital twin for real-world application. Source : [5]

1.1.2 Context

In this project, a digital twin acts as a precise and efficient method to simulate traffic based on real-world data. Digital twins leverage continuous data streams (for example, from vehicle counting devices) to create an up-to-date virtual model of physical traffic flows. Accurate traffic simulation relies on high-resolution data, typically gathered by a variety of counting devices such as inductive loop traffic detectors, Bluetooth recognition systems and ANPR (Automatic Number Plate Recognition) cameras [6]. These devices measure the number of vehicles passing specific locations over time, producing the data needed for dynamic modeling and simulation.

Through the aggregation and analysis of such data, it becomes possible to estimate traffic density and infer vehicle movement patterns across a road network. In this framework, physical traffic corresponds to the physical entity of the digital twin, real-time vehicle counts represent the virtual layer and the counting devices themselves serve as the communication link between the physical and digital components.

Traffic simulation is particularly vital in urban environments such as Brussels, where the frequent congestion and ongoing roadworks demand dynamic management of the transport network. Planned deviations due to construction may alter typical traffic flows and produce unexpected bottlenecks. Accurate simulation enables city planners to evaluate alternative deviation plans and identify those that minimize congestion and disruption. Additionally, the direct and indirect effects of traffic interventions, such as road closures, can be promptly assessed.

1.2 Problem statement

The thesis problem : Given a baseline traffic situation and a disruptive event (road closure) the goal is to generate feasible deviation plans and assess their impact on traffic in order to recommend the best plan to the urban planner.

1.2.1 Objective

The main objective of this work is the development of a web-based application that reproduces and extends the functionalities of an existing tool, TrafficTwin¹ [7]. This modernization enhances accessibility and usability. The work is divided into several sub-goals:

TO BE WRITTEN

Ultimately, the goal is to provide city planners and traffic managers with actionable, data-driven tools for congestion mitigation and urban mobility planning.

1.3 Research questions

Present the research questions that will be addressed in the thesis. The main one is “How deviation plans should be evaluated?” and the subquestions are “What are the key factors influencing the evaluation of deviation plans?” and “How can we develop a

¹<https://traffictwin.ulb.be>

decision-support approach to compare multiple deviation plans?”.

1.4 Contributions and thesis outline

Summarize the contributions (end-to-end deviation plan assessment pipeline, candidate plan generation strategies, Monte Carlo evaluation protocol). It ends with a short outline of the thesis chapters.

2 Foundations and Related Work

2.1 Traffic digital twins for decision support

Define the concept of a traffic digital twin in the context of urban traffic planning. It reviews how digital twins are used for analyses and positions the work specifically on disruption management and detour assessment.

2.1.1 Digital twins in traffic simulation

Digital twin and data mining. Recent advances have seen digital twin leveraged in urban mobility to integrate real-time and historical data (from sensors, cameras and vehicles) [8]. The integration of advanced data mining techniques such as trajectory mining, which exploits the information of trajectories for both private and public vehicles, and process mining, which extracts processes from available event logs in information systems, within digital twin frameworks can reveal mobility patterns, detect traffic anomalies, and simulate possible scenarios.

However, the deployment of such systems is not without significant challenges. Foremost among these is the need for continuous, reliable data acquisition. In addition, although mining trajectory data enhances the ability to simulate mobility behaviors, it also raises significant privacy issues. For large metropolitan areas, the demand for real-time processing introduces further complications, as scenario simulations often need to meet stringent latency constraints.

Digital twins in intelligent transport systems. Digital twins in intelligent transport systems (ITS) are digital replicas of real-world transport networks, providing a testbed for simulating, analyzing, and optimizing traffic management strategies before their deployment in the physical world [9]. ITS integrate services across the city (including traffic authorities, emergency responders, and public services) to enable greater coordination, faster response times, and improved interactions between road users compared to conventional systems. As discussed in [9], digital twins can model dynamics of urban traffic flows by integrating real-time data from sensors, vehicles, and control centers, thereby supporting predictive analytics and data-driven decision making.

These digital representations play a vital role in modern intelligent transportation systems, enabling traffic management centers to shift from operator-dependent decision-making to more automated and data-driven strategies. According to Rudskoy et al. [9], the definition of a digital twin in this context is “a module that reproduces a detailed digital model of the road and allows for modeling and experiments.” The primary purpose is to test solutions and simulate various traffic scenarios before implementation in the real world.

The implementation of digital twins in ITS helps address key challenges in transport networks including traffic optimization, management of transport during accidents or special events and improving road safety.

2.1.2 Simulation and mobility

Simulation plays a pivotal role in mobility research, enabling the testing and analysis of various traffic management and vehicle technologies in controlled, reproducible environments. Two notable applications from recent literature are highlighted below:

AI-driven traffic simulation. Talha Azfar and Ruimin Ke [10] have demonstrated the effectiveness of combining artificial intelligence with co-simulation environments for urban mobility. Co-simulation means running different simulators together to model how connected systems work as a whole [11]. AI-agents using computer vision (YOLO V5/V8², which are models designed for object detection) for real-time traffic observation were compared against statically configured traffic lights. Performance metrics such as mean speed, total travel time and mean waiting time were evaluated under medium and heavy traffic conditions. The results shows that the AI-agents reduce mean waiting time and total trip duration especially with ground-truth data.

These results underscore the importance of simulation-based optimization. The integration of multi-agent reinforcement learning and computer vision in a co-simulation framework enables realistic evaluation and adaptation of traffic control strategies before real-world deployment.

Simulation-based traffic light optimization. Mehdi et al. [12] proposed an innovative modification to the traditional three-phase traffic light system by introducing a four-stroke traffic light and directly measuring its impact at a critical intersection in downtown Urmia, Iran. While previous research has explored a variety of advanced algorithms (including queuing theory, stochastic modeling, variable-phase timing, and even neural networks) to minimize vehicle waiting times, this work focuses on an alternative signal configuration.

By using Arena Simulation Software³, the authors developed a comprehensive model capturing real-world vehicle arrival and departure patterns at the intersection. Through systematic comparison of the conventional three-phase setup with the proposed four-phase system across multiple traffic scenarios, the results indicated that the four-stroke signal configuration significantly reduced the average time vehicles spent in the system.

Although the study is centered on a single intersection, its findings highlight the potential benefits of rethinking conventional traffic light algorithms. Future research should investigate whether implementing such four-phase signal schemes at a broader, network-wide scale could further alleviate congestion and improve overall urban mobility.

2.1.3 Positioning of this work

This work builds upon and extends the several existing contributions discussed before in the field of traffic simulation and digital twins. A tutorial for traffic modeling using SUMO provides a step-by-step guide to generate and simulate traffic scenarios [13]. This methodology serves as the foundation for constructing a realistic simulation scenario for the Brussels region, contributing to the development of a reliable digital twin based on local data. Additionally, while a case study already exists for traffic simulation in

²<https://docs.ultralytics.com/fr/yolov5/>

³<https://www.rockwellautomation.com/fr-be/products/software/arena-simulation.html>

Brussels using incomplete data [14], the proposed approach further leverages the HybridIoT methodology introduced in that work to enhance model accuracy by estimating missing traffic information. HybridIoT is a multi-agent system (MAS)-based approach designed to estimate missing environmental data in large-scale environments [15]. Lastly, as discussed before, an interactive tool named *TrafficTwin* has been developed to support traffic management experts in designing and evaluating road deviation plans [7]. The current work builds on this concept, adapting and extending it. A new module is proposed for SUMO to facilitate the creation and simulation of detour strategies, with the aim of assisting authorities such as *Bruxelles Mobilité*⁴ in optimizing roadwork planning. The overarching goal is to ensure the tool remains accessible, with the potential for broader deployment to the general public.

2.2 Traffic simulation for operational planning

Introduce microscopic traffic simulation principles and explains why SUMO is appropriate for the work. It also discusses key limitations.

2.2.1 Traffic simulation tools

Traffic congestion is one of the major challenges faced by modern cities and is expected to get worse with increasing population and vehicle usage [16], [17]. Traffic simulation tools have been developed as mathematical and computational models that replicate real-world traffic behavior and dynamics [18]. These tools help transportation planners and engineers to optimize road networks designs or estimate traffic-related emissions.

Traffic simulations are commonly categorized into three main types based on the level of detail desired [19]:

1. **Microscopic Simulation:** Models each traffic participant (e.g., vehicles, pedestrians) individually. Agents continuously in time and space respond to their immediate environment. The global traffic dynamics is the result of the sum of these individual decisions and behaviors.
2. **Mesoscopic Simulation:** Still represents individual agents, but their behavior is determined by average traffic conditions like speed or density. Agents only interact directly at key points in the network, such as intersections.
3. **Macroscopic Simulation:** Looks at traffic as a whole rather than focusing on individual vehicles.

Choosing a simulation type entails balancing computational cost and realism. For instance, microscopic simulation is ideal for projects that require modeling of individual behaviors or the testing of interventions at the street or intersection level, which is often necessary for a dense urban environment such as Brussels. In contrast, a macroscopic simulation might suffice for citywide traffic forecasting or strategic planning, where the focus is on broader patterns rather than local details.

⁴<https://be.brussels/fr/propos-de-la-region/structure-et-organisation/administrations-et-institutions-de-la-region/bruxelles-mobilite>

2.2.2 SUMO

SUMO (Simulation of Urban MObility) is an open-source, highly portable, microscopic and continuous multimodal traffic simulation tool. It allows detailed modeling of individual vehicles, each following its own route through a specified road network. As a microscopic simulator, SUMO treats every vehicle as an individual entity with specific behaviors and interactions. The tool is multi-modal enabling the simulation of diverse traffic participants including cars, public transport vehicles and pedestrians [20]–[22].

SUMO provides some tools to setup the simulation and managing the data of the road network [20], [22]. In this project, the following components of the SUMO ecosystem are particularly relevant:

- **sumo**: Executes command-line-based microscopic traffic simulations without graphical visualization.
- **netconvert**: Imports and converts road network data from various formats into the internal SUMO network format.
- **duarouter**: Computes optimal routes across the network based on demand inputs and performs Dynamic User Assignment (DUA).

SUMO has been used extensively to model real-world traffic systems in diverse urban contexts, including cities such as Berlin, Dublin and Turin [23]. Each scenario adapts to the specific traffic dynamics and urban characteristics of its location:

1. **Berlin**: Berlin has served for large-scale traffic simulations using SUMO. The “24 Hours of Traffic in Berlin” scenario captures over 2.2 million trips within an 800km² area. This model enhances traditional traffic simulations by incorporating motorized private vehicles alongside realistic bus schedules derived from actual General Transit Feed Specification (GTFS) data, representing 89% of the city’s bus lines. This scenario allows for holistic analyses that combine traffic, communication (including V2X and LTE/5G) and application simulations, making it suited for smart mobility researches. [24]
2. **Dublin**: The study by Gueriau and Dusparic [25] leverages SUMO to simulate and analyze the impact of connected and autonomous vehicles (CAVs) on real Irish road networks, covering urban, national and motorway settings. Using detailed demand and network data from Dublin and surrounding areas, the research quantifies how varying CAV penetration rates affect both traffic efficiency and open dataset for benchmarking the effects of emerging vehicle technologies in mixed traffic environments.
3. **Turin**: Leveraging real-world data, the TuSTScenario [26] provides a highly detailed SUMO-based simulation of metropolitan Turin, covering over 602.61km². The model integrates structural road information from OpenStreetMap, validated average speeds from Google APIs, actual traffic light programming and real trip demands from local mobility services. With more than 2.2 million simulated daily trips and validated against city sensor data, this scenario enables realistic studies of urban traffic dynamics and congestion patterns.

Brussels is also a large city warranting similarly large-scale simulations. However, the

focus of this study is distinct: rather than prioritizing multi-modal integration or CAV impact, we specifically investigate the planning and management of roadworks within the Brussels network. The goal is to demonstrate how such interventions can be realistically modeled, analyzed, and optimized in SUMO for modern urban infrastructure management.

Beyond traditional traffic flow analysis SUMO has been used to evaluate the effects of technologies like virtual traffic lights communication systems where cars directly communicate with each other [27]. This study showed that replacing traffic lights with virtual ones can reduce both congestion and the likelihood of traffic accidents. Moreover SUMO can be used in environmental research, for instance in assessing the impact of traffic-related emissions on urban air quality [28].

One of the strengths of SUMO is its deterministic core which allows reproducibility of experiments. However, this deterministic nature can be a limitation when it comes to capturing the variability in human driving behavior which requires the integration of stochastic elements [29].

2.2.3 Alternatives to SUMO

Several other traffic simulation tools can serve as alternatives to SUMO, each with different features, advantages and limitations:

1. **PTV Vissim**: PTV-Vissim⁵ is a widely used microscopic traffic simulation software that also supports multi-modal transport systems. It offers a broad set of features comparable to those of SUMO. However, it is a proprietary tool that requires a commercial license, which can restrict its accessibility and limit the ability to implement custom modifications at the code level.
2. **Aimsun**: Aimsun⁶ is a commercial traffic simulation platform that supports microscopic, mesoscopic, and macroscopic modeling. While it is used in various contexts, its commercial nature introduces licensing costs and reduces flexibility for developers who require deeper code-level access and customization.
3. **CityFlow**: CityFlow [30] an open-source microscopic simulator designed for efficient, large-scale urban traffic simulations, capable of handling thousands of intersections with high computational efficiency. Its architecture is highly optimized for speed and parallel execution, making it over twenty times faster than SUMO in large-scale benchmarks [31]. CityFlow is particularly popular in the field of artificial intelligence and reinforcement learning, serving as a benchmark for multi-agent adaptive traffic signal control and dynamic routing tasks. However, CityFlow remains limited compared to SUMO in certain areas: it offers fewer out-of-the-box features for conventional traffic simulation, has a less mature ecosystem, and does not natively support the simulation of public transportation modes such as buses, trams, or trains, which restricts its suitability for comprehensive, multi-modal urban simulations.

⁵<https://www.ptvgroup.com/en/products/ptv-vissim>

⁶<https://docs.aimsun.com/>

2.2.4 Why prefer SUMO

SUMO was selected for this project based on the following criteria:

- **Open-source:** SUMO’s open-source status allows users to freely modify, adapt, and extend the software, making it suitable for research and academic use where flexibility and transparency are important. This degree of code-level access is not available in proprietary tools like PTV Vissim or Aimsun.
- **Community and documentation:** SUMO benefits from a large active community that contributes regularly to new features or bug fixes. In addition, it offers extensive documentation and a wide range of tutorials, which significantly lower the learning curve for new users.
- **Microscopic and multi-modal simulation:** SUMO provides detailed agent-based (microscopic) modeling of cars, public transport, and pedestrians. While other simulators such as Vissim and Aimsun also support microscopic and multi-modal simulation, SUMO’s implementation is open, which enables deeper customization and integration for research needs.
- **Integration and flexibility:** SUMO’s design and command-line tools (like `netconvert` and `duarouter`) facilitates working on various data sources making it easy to connect with external components.

Commercial tools like PTV Vissim and Aimsun offer advanced graphical user interfaces and support; however, their proprietary nature limits customizability and long-term usability in research settings. CityFlow, while open-source and scalable, lacks SUMO’s extensive documentation and wider range of traffic simulation features relevant for this project’s objectives.

2.3 Metrics for urban traffic planning

Explain what “impact” is, it can be defined at different levels, network efficiency, specific user impact, delay etc. Motivates the need of a multi-metric perspective and sets the foundations for later scoring and ranking strategies

2.4 Stochastic simulation

Why a single run is not enough, introduce Monte Carlo simulation, estimating performance distributions and supporting decisions with confidence.

In the context of a Digital Twin for traffic management (especially one designed to evaluate deviation plans for disruptions like roadworks or accidents) the treatment of uncertainty is not merely a technical refinement but a fundamental requirement for validity [32], [33]. Traffic is a stochastic process and small perturbations in the initial conditions can propagate to cause macroscopic differences in flow and congestion patterns [34].

Stochastic simulation introduces randomness into the model to mimic the real-world uncertainties in demand, driver behaviour and network conditions [35]. By treating model inputs and parameters as random variables rather than fixed values, stochastic

simulations generate a distribution of possible outcomes that better reflects the variability of traffic states. This transition from point-estimates to probabilistic distributions allows for the application of risk assessment frameworks, enabling planners to weigh the trade-offs between efficiency and reliability.

2.4.1 Theoretical foundations of the Monte Carlo method

The Monte Carlo method is a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. The underlying concept is to use randomness to solve problems that might be deterministic in principle but are too complex to solve analytically due to the high dimensionality of the problem.

2.5 Multiple-criteria decision analysis

The multiple-criteria decision analysis (MCDA) aims to help in situations where multiple criteria, most of the time contradictory, must be.

3 Methodology

3.1 System overview and pipeline

Present the system overview and the pipeline of the thesis. It should be clear how the digital twin is used to generate candidate plans and how the Monte Carlo evaluation protocol is used to assess them.

The project is organized in four different main modules, each one corresponding to one step of the pipeline :

Module	Main role	Main outputs
scenario_generator	Generates synthetic traffic demand and initial routes	routes.xml, OD matrices
deviation_plan_maker	Converts the network into a graph and calculates the deviation plan(s)	JSON of candidates deviation plans
monte_carlo_simulation	SUMO simulation with multiple runs with random seeds	run_# folders, summary.json
evaluator	Statistical analysis and multi-criteria ranking	evaluation.json

Table 1: Functional organization of the pipeline

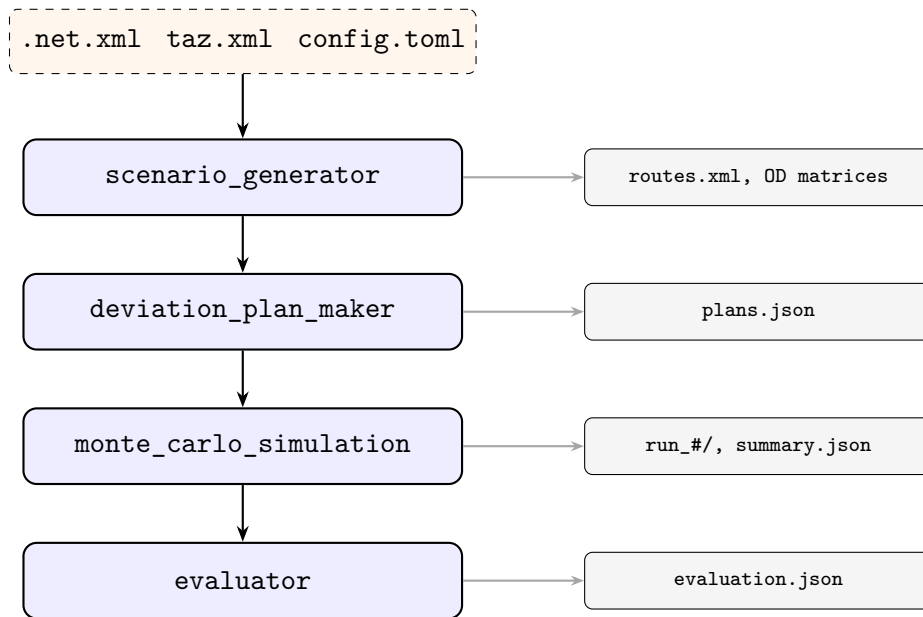


Figure 3: Four-module pipeline. Inputs (dashed box) flow top-to-bottom through the four modules. Each module’s main output artefacts are shown to the right.

3.2 Scenario generation

The role of the scenario generator is to create demand that is :

- **network-aware** because routes are built on the real road network
- **zone-aware** because demand is expressed at the TAZ level
- **temporally structured** because daily traffic should exhibit morning and evening peaks
- **directionally realistic** because peak periods should concentrate trips from residential areas toward employment centers and back.

3.2.1 Module overview

Module	Main role
<code>generator.py</code>	command-line entry point, loads TOML config and launches generation
<code>config_manager.py</code>	typed access to configuration values and path resolution
<code>zones.py</code>	TAZ loading, edge filtering, zone mass and endpoint sampling
<code>od_model.py</code>	OD model, temporal profile
<code>route_library.py</code>	TAZ-to-TAZ impedance estimation and candidate-route generation with caching
<code>vehicle_sampler.py</code>	conversion from hourly OD cells to individual vehicles with route choice
<code>exports.py</code>	writing <code>routes.xml</code> , <code>out_od.xml</code> and hourly CSV matrices
<code>traffic_generator.py</code>	high-level orchestration of the three pipeline stages

Table 2: Structure of the `scenario_generator` package.

Table 2 summarizes the main concepts of the `scenario_generator`. The generator takes as input :

- a SUMO road network (`.net.xml`),
- a Traffic Analysis Zone (TAZ) file (`taz.xml`),
- a TOML configuration file specifying demand and routing parameters.

It produces:

- a SUMO vehicle route file (`routes.xml`) that can be used directly as a demand input to `sumo` or `sumo-gui`,
- hourly OD matrices in both XML (`out_od.xml`) and CSV form for inspection and re-use.

The pipeline consists of three sequential stages:

1. **Stage 1 - Demand generation:** construct 24 hourly Origin–Destination (OD) matrices using a gravity model with an optional directional commute overlay.

2. **Stage 2 - Route library:** build a small set of candidate routes for each active OD pair using Yen’s k -shortest paths algorithm [36].
3. **Stage 3 - Vehicle instantiation:** assign a departure time and a concrete route to every trip.

3.2.2 Stage 1 : Demand generation

3.2.2.1 Traffic Analysis Zones

The spatial unit of the model is the *Traffic Analysis Zone* (TAZ), a standard construct in transport demand modelling that aggregates individual trip origins and destinations into discrete geographic areas. TAZs in SUMO are defined as polygons that can be associated with the road network. Each TAZ is defined by the set of SUMO road edges it contains.

A scalar *zone mass* m_i is assigned to each TAZ as the capacity of its drivable edge lengths :

$$m_i = \sum_{e \in \mathcal{E}_i} \ell(e) \cdot n(e) \quad (1)$$

where $\ell(e)$ is the length of edge e in meters and $n(e)$ is its lane count.

Equation 1 is a *deliberate approximation of the network structure* adopted in the absence of socio-economic calibration data. In the classical four-step transport model, trips productions and attractions are estimated using socio-economic data [37] but none of those are available in the present synthetic setting. Lane kilometers⁷ are used here as a surrogate because they are a directly measurable network attribute and because major road corridors tend to account for a disproportionate share of traffic volume relative to their share of network length. The resulting m_i has therefore to be replaced by observed productions and attractions whenever such data become available.

3.2.2.2 Temporal Demand Profile

The total daily demand N is distributed across the 24 hours of the simulation using a *two-Gaussian profile*:

$$p_h = \frac{a_m g(h; \mu_m, \sigma) + a_e g(h; \mu_e, \sigma)}{\sum_{h'=0}^{23} [a_m g(h'; \mu_m, \sigma) + a_e g(h'; \mu_e, \sigma)]}, \quad (2)$$

where

$$g(h; \mu, \sigma) = \exp\left(-\frac{(h - \mu)^2}{2\sigma^2}\right), \quad (3)$$

μ_m and μ_e are the morning and evening peak hours respectively, and σ controls the temporal spread. The amplitudes a_m and a_e allow the two peaks to differ in magnitude, the default configuration sets $a_e > a_m$ to reflect the slightly larger afternoon peak typically observed in urban network [38].

The continuous profile p_h is converted into hourly trip counts n_h using the largest-remainder method which guarantees $\sum_h n_h = N$ while preserving profile proportions as closely as possible under the integer constraint.

⁷A lane kilometer is a unit of length that is equal to one kilometer of road with one lane.

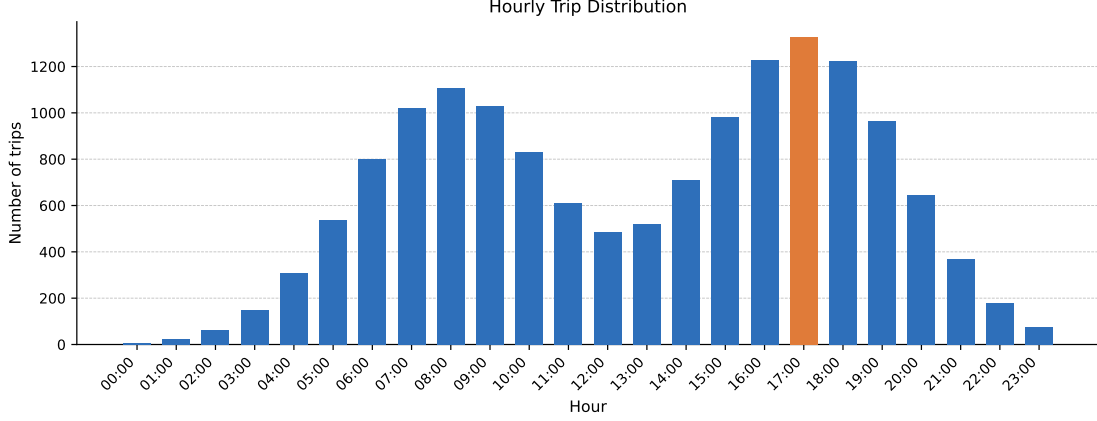


Figure 4: Hourly trip distribution produced by the two-Gaussian profile (Equation 2) for the test scenario. The orange bar marks the evening peak, which is slightly larger than the morning peak ($a_e > a_m$).

3.2.2.3 Gravity OD model

The spatial distribution of trips across TAZ pairs is governed by a *singly-constrained gravity model* [39]. The **unnormalized** weight of the OD pair (i, j) is :

$$w_{ij} = m_i \cdot m_j \cdot \exp(-\beta c_{ij}), \quad i \neq j, \quad (4)$$

where c_{ij} is the travel-cost impedance between zones i and j (shortest path length in meters) and $\beta \geq 0$ is the distance decay parameter⁸.

Pairs with infinite or negative impedance are assigned to zero weight. The **normalized OD share matrix** is obtained as :

$$S_{ij} = \underbrace{\frac{m_i}{\sum_k m_k}}_{\text{origin share } O_i} \cdot \underbrace{\frac{m_j \exp(-\beta c_{ij})}{\sum_\ell m_\ell \exp(-\beta c_{i\ell})}}_{\text{conditional destination probability } p(j|i)}, \quad (5)$$

so that, by construction, $\sum_j S_{ij} = m_i / \sum_k m_k$ and $\sum_{ij} S_{ij} = 1$.

For each hour h , the share matrix is scaled to the corresponding integer total n_h via the largest-remainder method, yielding an exact integer OD matrix T^h with $\sum_{ij} T_{ij}^h = n_h$.

The model is *singly-constrained on the production side*. Although the weight $w_{ij} = m_i m_j \exp(-\beta c_{ij})$ depends symmetrically on origin and destination masses, the normalisation in Equation 5 enforces the row-sum constraint

$$\sum_j S_{ij} = \frac{m_i}{\sum_k m_k}$$

while leaving the column sums $\sum_i S_{ij}$ unconstrained. In Wilson's balancing-factor formulation **ortuzar_modelling_transport**, this corresponds to enforcing only the origin-side factor A_i and fixing the destination-side factor $B_j \equiv 1$. A doubly-constrained model would

⁸A larger β means trips decay faster with distance/cost (people are more reluctant to travel far), a smaller β means the distribution is more spatially spread out. Wilson discusses this in the context of calibrating the model to observed data.[39]

additionally enforce observed zonal attractions and would require both balancing factors to be recovered by iterative proportional fitting; zonal attractions are not observed in the present synthetic setting, so enforcing them would introduce unidentified parameters. The singly-constrained formulation is therefore the parsimonious choice that uses only the information actually contained in the network (zone masses and impedances) and leaves destination totals free to respond to the network structure through $\exp(-\beta c_{ij})$.

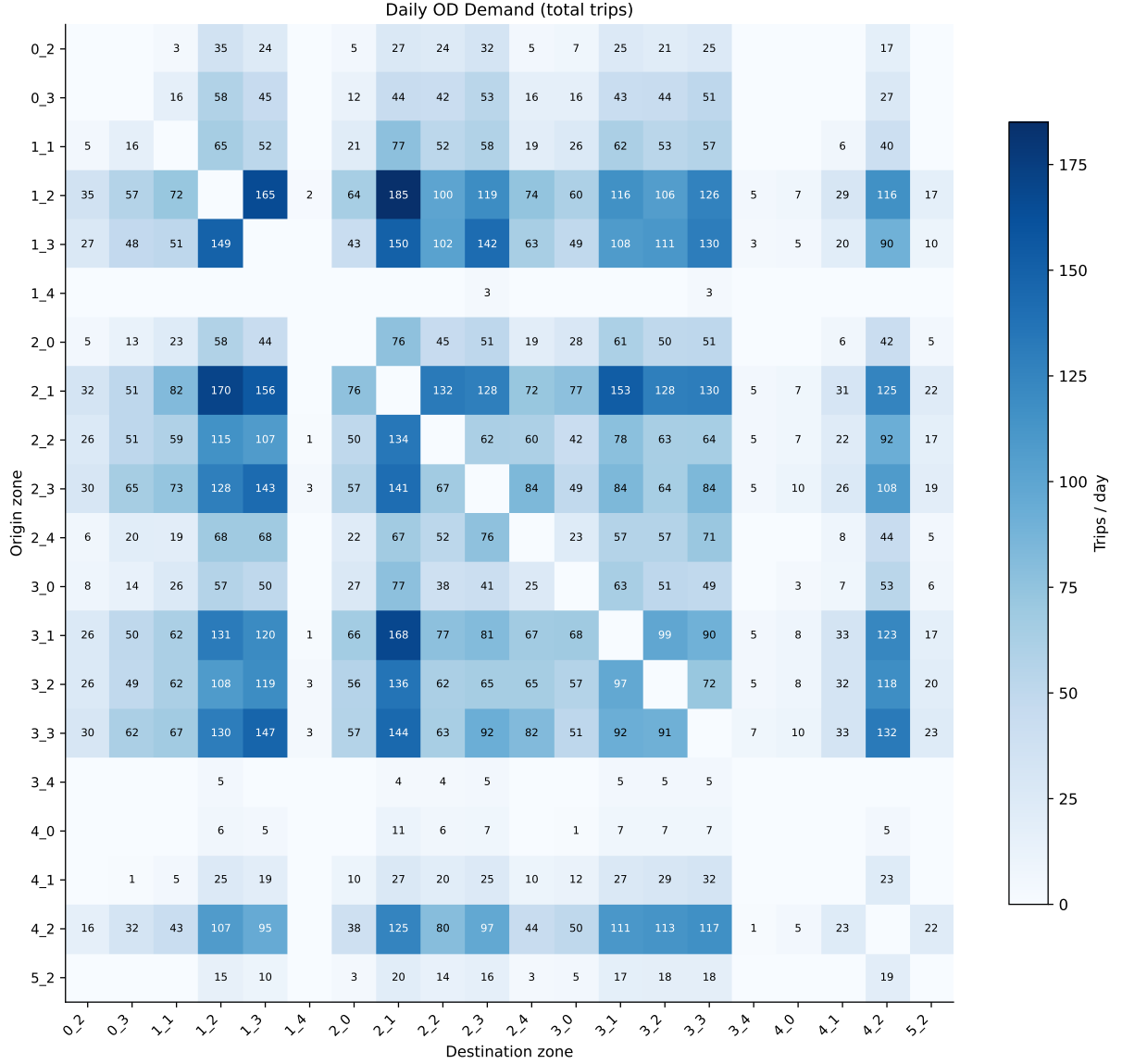


Figure 5: Daily OD demand matrix $\sum_h T^h$ for the test scenario (20 TAZs, 10 000 daily trips). Rows are origin zones, columns are destination zones. The cell colour and annotation encode the total number of trips. The diagonal is zero by construction (no intra demand). Dense clusters correspond to adjacent zones with high lane-kilometer mass.

3.2.2.4 Commute Overlay (Optional)

When residential weights r_i and employment weights e_i are provided for each zone, a directional commute bias is introduced into the hourly OD shares. Two commute matrices

are constructed following the same gravity structure as Equation 4: one for the morning peak, capturing residential-to-employment flows, and one for the evening peak, capturing the reverse direction:

$$w_{ij}^{\text{AM}} = m_i m_j f(c_{ij}) r_i e_j, \quad (6)$$

$$w_{ij}^{\text{PM}} = m_i m_j f(c_{ij}) e_i r_j, \quad (7)$$

where $f(c_{ij}) = \exp(-\beta_c c_{ij})$ is a distance-decay function that discounts longer or costlier trips. In the AM matrix, the additional factors r_i and e_j amplify flows leaving residential zones and arriving at employment zones, the PM matrix reverses this logic by weighting departures from employment zones (e_i) toward residential destinations (r_j).

These directional matrices are then blended with the base OD share matrix through a time-varying mixture. For each hour h , the blended share is:

$$S_{ij}^h \propto S_{ij}^{\text{base}} + \lambda g(h; \mu_m, \sigma) S_{ij}^{\text{AM}} + \lambda g(h; \mu_e, \sigma) S_{ij}^{\text{PM}}, \quad (8)$$

where $g(h; \mu, \sigma)$ is a Gaussian kernel centred on hour μ with spread σ , and λ (**peak_weight**) governs the overall strength of the commute signal relative to the background. The AM component peaks around the morning rush hour μ_m and fades outside it, the PM component does the same around the evening rush hour μ_e . Setting $\lambda = 0$ reduces the blended share to the base model alone. The resulting shares are normalised before being applied to the trip scaling step.

3.2.3 Stage 2 : Route library

3.2.3.1 Edge expanded network

Routing is performed on an *edge-expanded* directed graph in which SUMO road edges serve as nodes. A directed arc connects edge u to edge v whenever v is reachable from u through a SUMO connection, with arc weight equal to the length of v , path cost is therefore the sum of edge lengths in metres.

The edge-expanded representation is preferred to a junction-level (node-based) graph for two reasons specific to this pipeline. First, SUMO's demand format (**routes.xml**) requires routes to be expressed as ordered edge sequences, an edge-level graph produces paths that can be serialised directly, whereas a node-level graph would require an additional junction-to-edge conversion. Second, the edge-expanded graph natively encodes turning feasibility, an arc (u, v) exists if and only if SUMO permits the corresponding connection, so every path returned by the router is guaranteed to be physically drivable without any posterior feasibility check. Internal junction edges ⁹ and edges that do not permit private-vehicle access are excluded at graph construction time.

3.2.3.2 Impedance Estimation

The gravity model in Stage 1 requires a TAZ-to-TAZ impedance matrix. Computing exact center-to-center shortest paths would require all-pairs Dijkstra on a large graph, which is unnecessarily expensive for the present purpose. Instead, for each directed zone pair

⁹SUMO ids beginning with :

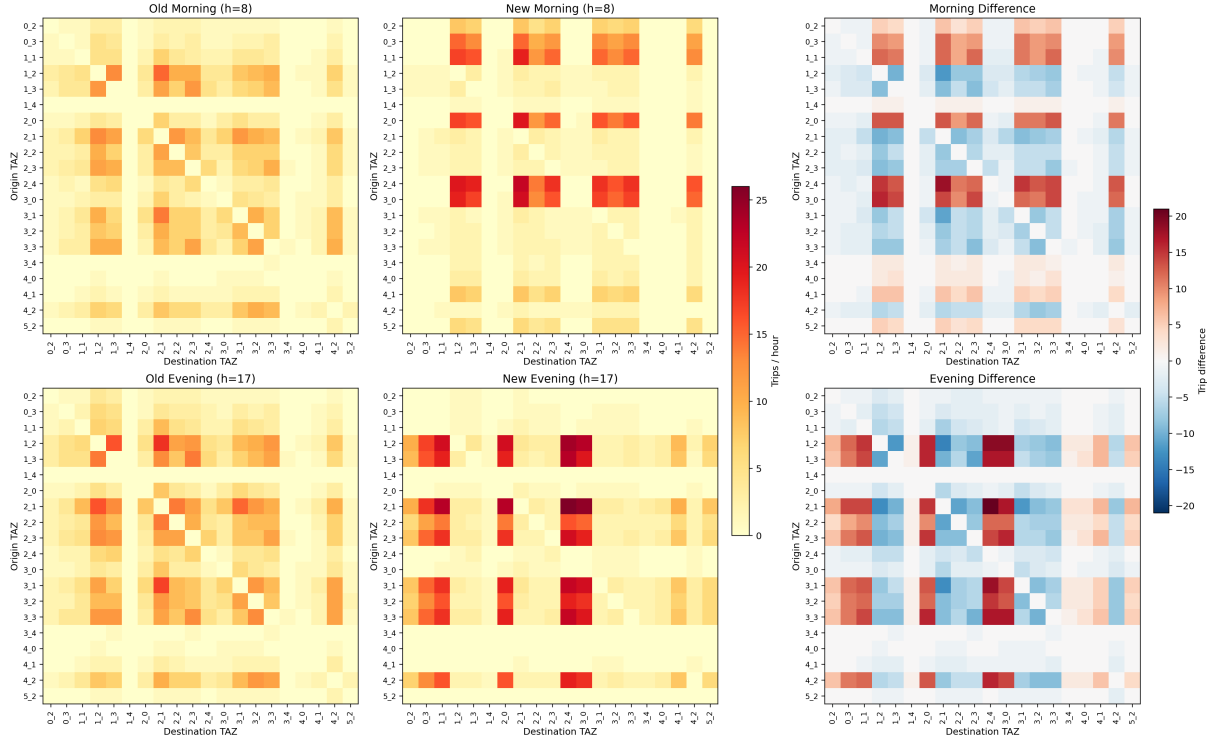


Figure 6: Effect of the commute overlay on the hourly OD matrices. Left column: base gravity model ($\lambda = 0$). Centre column: model with commute overlay ($\lambda > 0$). Right column: difference (centre – left). Top row: morning peak hour ($h = 8$). Bottom row: evening peak hour ($h = 17$). Red cells in the difference maps indicate flows amplified by the overlay, by blue cells indicate flows attenuated. The AM overlay concentrates demand on residential→employment corridors, the PM overlay reverses this pattern.

(i, j) , a small number of (origin-edge, destination-edge) combinations is sampled from the two zones and the minimum observed single-source shortest-path cost is taken as the representative impedance. The approximation is decreasing in geographic proximity, which is sufficient for the gravity model to produce a spatially plausible OD distribution.

3.2.3.3 k -Shortest Candidate Routes

For each active OD pair (i.e. $T_{ij}^h > 0$ for at least one hour h), a set of at most k candidate routes is computed as follows:

1. Sample s concrete (origin-edge, destination-edge) pairs, where edges are drawn proportionally to zone mass (Equation (1)).
2. For each pair, enumerate up to k shortest simple paths using Yen’s algorithm [36]
3. Deduplicate across all samples by edge sequence, if the same sequence appears more than once, retain the lowest-cost instance.
4. Return at most k candidates sorted by cost.

3.2.4 Stage 3 : Vehicle Instantiation

Once the hourly OD matrices and the candidate route library are available, every trip in the demand model is turned into a concrete SUMO vehicle definition. Each trip requires two random draws: a departure time and a route.

3.2.4.1 Departure Time

Trips within each one-hour interval $[b, b + 3600)$ are assumed to depart uniformly over that interval. For an OD cell containing n trips, departure times are drawn independently as:

$$t_1, \dots, t_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(b, b + 3600). \quad (9)$$

This is the natural maximum-entropy assumption when only hourly totals are known: given no information about the exact timing of individual trips within an hour, all departure instants in $[b, b + 3600)$ are treated as equally likely.

3.2.4.2 Route Selection

Each trip is assigned one route selected *uniformly at random* from the k precomputed candidates for its OD pair:

$$P(r \mid \mathcal{R}) = \frac{1}{|\mathcal{R}|}, \quad r \in \mathcal{R}. \quad (10)$$

This is again the maximum-entropy choice in the absence of driver-preference data: if no reason is given to favour one candidate over another, all candidates are equally likely. The approach ensures that, on average, the generated traffic load is spread across the full set of precomputed alternatives rather than being concentrated on a single route per OD pair.

The two draws (departure time and route) are performed with a single shared random-number generator whose seed is fixed in the configuration file, making the full scenario exactly reproducible.

3.2.5 Summary of Parameters

Table 3: Key parameters of the scenario generator and their roles.

Parameter	Stage	Effect
daily_total	1	Total daily trip count N
distance_decay_beta β	1	Gravity model decay; higher $\Rightarrow$ shorter trips
morning_peak_hour μ_m	1	AM peak centre (hours)
evening_peak_hour μ_e	1	PM peak centre (hours)
sigma_hours σ	1	Temporal spread of peaks
peak_weight λ	1	Commute overlay strength
k_paths k	2	Number of candidate routes per OD pair
endpoint_samples_per_od s	1 & 2	Accuracy vs. runtime trade-off

4 Implementation

4.1 Baseline prototype description

Present the original prototype description.

4.2 Identified limitations

Say what the limitations are in the context of the thesis, what we need to change.

4.3 Implemented improvements

Present the implemented improvements.

5 Results

5.1 Scenario set

The disruption scenarios used for evaluation (e.g., Annie Cordy tunnel closure and additional closures), including the affected infrastructure and motivation for selecting them. It explains how scenarios differ and why they collectively provide a meaningful testbed for the proposed methodology.

5.1.1 Use cases

This section highlights meaningful and realistic use cases where the simulation tool could be applied. These scenarios are particularly relevant in the context of planned road closures and the design of deviation plans. By simulating these situations, the tool can help anticipate traffic behavior. Below are several examples, with a focus on the Brussels region, that illustrate the potential of such simulations for urban traffic management:

1. **Roadworks at Carrefour Léonard:** The Carrefour Léonard is a well-known place where big roadworks are happening. Simulating such a scenario could be highly valuable as the tool could be used to compare multiple deviation strategies and identify the one that best reduces congestion.
2. **Closure of a major tunnel:** Tunnel closures are not uncommon in Brussels, whether due to maintenance or unexpected incidents. While the immediate effects on adjacent streets are often visible the simulation tool could provide insights into other consequences, such as congestion in distant neighborhoods. This is especially relevant in the age of GPS navigation systems where drivers may be rerouted through less obvious paths, potentially overloading streets not originally designed for heavy traffic.
3. **Closure of a neighborhood:** The complete closure of a neighborhood (for a marathon, festival...) presents complex traffic challenges. Not only does it block local roads but it also changes the flow of vehicles in the surrounding areas. With the simulation tool, traffic planners could evaluate different deviation strategies to minimize congestion.

5.2 Experimental setup per scenario

Present the experimental setup per scenario. It should be clear how the tool is used to generate candidate plans and how the Monte Carlo evaluation protocol is used to assess them.

5.2.1 Scenario 1 : Grid toy

5.3 Results presentation

Present the results of the experiments.

6 Discussion

6.1 Key findings

Present the key findings of the thesis.

6.2 Practical implications for urban planners

Present the practical implications for urban planners.

6.3 Limitations and validity checks

This section critically discusses limitations (including simulation realism) and frames these as threats to validity and explains how they might bias the evaluation or restrict generalization of findings.

6.4 Future work

Present the most impactful next steps for the work.

7 Conclusion

Some articles to read

- Reliability and Unreliability Aspects of Travel Time Analysis on the Stochastic Network Using the Target-Oriented Perspective - MDPI, , <https://www.mdpi.com/2071-1050/16/12/5148>
- Travel Time and Weather-Aware Traffic Forecasting in a Conformal Graph Neural Network Framework - arXiv, , <https://arxiv.org/html/2509.12043>
- Monte Carlo-Driven Decision Support for Subsea Pipeline Decommissioning Under Uncertainty - ASME Digital Collection, , <https://asmedigitalcollection.asme.org/offshoremechanics/article/148/1/011401/1219574/Monte-Carlo-Driven-Decision-Support-for-Subsea>
- 2005: Risk Management in Supply Networks Using Monte-Carlo Simulation, , <https://informs-sim.org/wsc05papers/202.pdf>
- Monte Carlo Simulation | SKYbrary Aviation Safety, , <https://skybrary.aero/articles/monte-carlo-simulation>
- Monte Carlo method - Wikipedia, , https://en.wikipedia.org/wiki/Monte_Carlo_method
- Bootstrap confidence intervals - how many replications to choose? - Stats StackExchange, , <https://stats.stackexchange.com/questions/357548/bootstrap-confidence-intervals-how-many-replications-to-choose>
- Quantitative Methodology for Comparing Microscopic Traffic Simulators - MDPI, , <https://www.mdpi.com/2673-7590/5/4/201>
- Simulation Modeling and Analysis / A.M. Law, W.D. Kelton. | Request PDF - ResearchGate, , https://www.researchgate.net/publication/31639535_Simulation_Modeling_and_Analysis_AM_Law_WD_Kelton
- End-To-End Uncertainty Quantification with Analytical Derivatives for Design Under Uncertainty, , https://ntrs.nasa.gov/api/citations/20230017353/downloads/Phillips_SciTech24.pdf
- Estimating confidence intervals for transport Mode share. - White Rose Research Online, , https://eprints.whiterose.ac.uk/id/eprint/2543/1/ITS2207-Estimating_Confidence_uploadable.pdf
- Randomness - SUMO Documentation - Eclipse SUMO - Simulation of Urban MObility, , <https://sumo.dlr.de/docs/Simulation/Randomness.html>
- Traffic Modeling with SUMO: a Tutorial - arXiv, , <https://arxiv.org/html/2304.05982v2>
- Randomness - SUMO wiki ja, , <https://kudzuyu.github.io/SUMO-wiki-ja/Simulation/Randomness/>

- Re: [sumo-user] Insert Randomness on route file - Eclipse Foundation, , <https://www.eclipse.org/lists/sumo-user/msg01346.html>
- Simulation-based travel time reliable signal control - MIT, , http://web.mit.edu/osorio/c/www/papers/osoCheSan_reliableSO.pdf
- A note on the number of replication runs in stochastic traffic simulation models, , https://www.researchgate.net/publication/237237852_A_note_on_the_number_of_replication_runs_in_stochastic_traffic_simulation_models
- Random seeds in transport models - Aimsun, , <https://www.aimsun.com/technical-notes/random-seed/>
- Mode Choice in Agent-Based Models: Efficiency and Stability - Research Collection, , <https://www.research-collection.ethz.ch/bitstreams/e173ee2b-762a-4ef8-950e-fdf08541efdf/download>
- 13.7.2 Number of Simulation Runs, , <https://www.txdot.gov/manuals/des/tsp/chapter-13-microsimulation-analysis/13-7-alternatives-analysis/13-7-2-number-of-simulation-runs.html>
- Guidelines for Applying Traffic Microsimulation Modeling Software 2019 Update to the 2004 Version - Chapter 6. Alternatives Analysis - FHWA Office of Operations, , <https://ops.fhwa.dot.gov/publications/fhwahop18036/chapter6.htm>
- Traffic Analysis Toolbox Volume III: Guidelines for Applying Traffic Microsimulation Modeling Software - Federal Highway Administration, , <https://highways.dot.gov/media/6916>
- appendix a. suggested number of scenarios and number of runs - FHWA Operations, , https://ops.fhwa.dot.gov/publications/fhwahop20057/app_a.htm
- Computer simulation modeling of recreation use: current status, case studies, and future directions - USDA Forest Service, , https://www.fs.usda.gov/rm/pubs/rmrs_gtr143.pdf
- Repeat a simulation with different seed values in SUMO - Stack Overflow, , <https://stackoverflow.com/questions/55214253/repeat-a-simulation-with-different-seed-values-in-sumo>
- Monte Carlo with Confidence Intervals - Noorjax Consulting, , <https://noorjax.com/2024/07/14/monte-carlo-with-confidence-intervals/>
- Batched Monte Carlo and the Confidence Interval, , <https://epubs.ans.org/download/?a=46594>
- Chapter: Appendix C: Reproducibility and Validity, , <https://www.nationalacademies.org/read/25335/chapter/18>
- Visualizing samples with box plots - Northeastern University, , https://course.khoury.northeastern.edu/cs7280sp16/CS7280-Spring16_files/NatureMethods-boxplots.pdf
- 15 traffic simulation models - Oregon.gov, , https://www.oregon.gov/odot/Planning/Documents/APMv2_Ch15.pdf

- Number of Replications Required in Monte Carlo Simulation Studies: A Synthesis of Four Studies - ResearchGate, , https://www.researchgate.net/publication/289453990_Number_of_Replications_Required_in_Monte_Carlo_Simulation_Studies_A_Synthesis_of_Four_Studies
- 4.2 Visualizations for Numeric Data: Exploring Train Ridership Data | Feature Engineering and Selection: A Practical Approach for Predictive Models - Bookdown, , <https://bookdown.org/max/FES/visualizations-for-numeric-data-exploring-train-ridership-data.html>
- Visual presentation of data by means of box plots | Request PDF - ResearchGate, , https://www.researchgate.net/publication/281121791_Visual_presentation_of_data_by_means_of_box_plots
- Box & Whisker Plot: Definition, Examples + [Excel Tutorial] - Formplus, , <https://www.formpl.us/blog/box-whisker-plot-definition-examples-excel-tutorial>
- Generalization of Boxplot for Ensembles of Curves - UC Davis Center for Visualization, , https://vis.cs.ucdavis.edu/vis2014papers/TVCG/papers/2654_20tvcg12-mirzargar-2346455.pdf
- Contour Boxplots: A Method for Characterizing Uncertainty in Feature Sets from Simulation Ensembles | Request PDF - ResearchGate, , https://www.researchgate.net/publication/256837300_Contour_Boxplots_A_Method_for_Characterizing_Uncertainty_in_Feature_Sets_from_Simulation_Ensembles
- Calibrating microscopic traffic models with macroscopic data - arXiv, , <https://arxiv.org/html/2409.19090v1>
- The Calibration of Traffic Simulation Models - JRC Publications Repository, , <https://publications.jrc.ec.europa.eu/repository/bitstream/JRC68403/lbna25188enn.pdf>

Bibliography

- [1] D. T. consortium. “Definition of a digital twin.” (2025), [Online]. Available: <https://www.digitaltwinconsortium.org/initiatives/the-definition-of-a-digital-twin/> (visited on 04/28/2025).
- [2] IBM. “What is a digital twin?” (2021), [Online]. Available: <https://www.ibm.com/think/topics/what-is-a-digital-twin> (visited on 04/28/2025).
- [3] E. Katsoulakis, Q. Wang, H. Wu, *et al.*, “Digital twins for health: A scoping review,” *npj Digital Medicine*, vol. 7, no. 1, p. 77, 2024.
- [4] D. Jones, C. Snider, A. Nassehi, J. Yon, and B. Hicks, “Characterising the digital twin: A systematic literature review,” *CIRP Journal of Manufacturing Science and Technology*, vol. 29, pp. 36–52, 2020, ISSN: 1755-5817. DOI: <https://doi.org/10.1016/j.cirpj.2020.02.002>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1755581720300110>.
- [5] A. Schaffland, J. Nelson, and J. Schöning, “Simulating traffic networks: Driving sumo towards digital twins,” *SUMO Conference Proceedings*, vol. 5, pp. 113–125, Jul. 2024. DOI: 10.52825/scp.v5i.1105. [Online]. Available: <https://www.tib-op.org/ojs/index.php/scp/article/view/1105>.
- [6] S. Hadavi, H. Buldeo Rai, S. Verlinde, H. Huang, C. Macharis, and T. Guns, “Analyzing passenger and freight vehicle movements from automatic-number plate recognition camera data,” English, *European Transport Research Review*, vol. 12, no. 1, pp. 1–17, May 2020, ISSN: 1867-0717. DOI: 10.1186/s12544-020-00405-x.
- [7] A. G. Davide, S.-M. Moisés, M.-P. Eladio, and B. Gianluca, *A simulation tool to assess the impact of deviation plans on disruptive events of urban traffic*.
- [8] A. Anniciello, S. Fioretto, E. Masciari, and E. V. Napolitano, “Digital twins for traffic congestion in smart cities: A novel solution using data mining techniques,” in *KMIS*, 2023, pp. 241–248.
- [9] A. Rudskoy, I. Ilin, and A. Prokhorov, “Digital twins in the intelligent transport systems,” *Transportation Research Procedia*, vol. 54, pp. 927–935, 2021.
- [10] T. Azfar and R. Ke, *Traffic co-simulation framework empowered by infrastructure camera sensing and reinforcement learning*, 2024. arXiv: 2412.03925 [eess.SY]. [Online]. Available: <https://arxiv.org/abs/2412.03925>.
- [11] C. Gomes, C. Thule, D. Broman, P. G. Larsen, and H. Vangheluwe, “Co-simulation: A survey,” *ACM Computing Surveys (CSUR)*, vol. 51, no. 3, pp. 1–33, 2018.
- [12] M. A. Kamran, H. Ramezani, S. Masoumzadeh, and F. Nikkhoo, “Traffic light signal timing using simulation,” *Communications on Advanced Computational Science with Applications*, vol. 2017, pp. 1–11, Jan. 2017. DOI: 10.5899/2017/cacsa-00061.
- [13] D. A. Guastella, E. Montero-Porras, A. Morales-Hernández, and G. Bontempi, *Traffic modeling with sumo: A tutorial*, 2025. arXiv: 2304.05982 [cs.NI]. [Online]. Available: <https://arxiv.org/abs/2304.05982>.

- [14] D. A. Guastella, B. Cornelis, and G. Bontempi, “Traffic simulation with incomplete data: The case of brussels,” in *Proceedings of the 1st ACM SIGSPATIAL International Workshop on Methods for Enriched Mobility Data: Emerging Issues and Ethical Perspectives 2023*, ser. EMODE ’23, Hamburg, Germany: Association for Computing Machinery, 2023, pp. 15–24, ISBN: 9798400703478. DOI: 10.1145/3615885.3628004. [Online]. Available: <https://doi.org/10.1145/3615885.3628004>.
- [15] D. A. Guastella, V. Campss, and M.-P. Gleizes, “A cooperative multi-agent system for crowd sensing based estimation in smart cities,” *IEEE Access*, vol. 8, pp. 183 051–183 070, 2020.
- [16] N. Serok, S. Havlin, and E. Blumenfeld Lieberthal, “Identification, cost evaluation, and prioritization of urban traffic congestions and their origin,” *Scientific Reports*, vol. 12, no. 1, p. 13 026, 2022.
- [17] S. Çolak *et al.*, “Understanding congested travel in urban areas,” *Nature Communications*, vol. 7, p. 10 793, 2016. DOI: 10.1038/ncomms10793.
- [18] Priyanka. “Top 5 leaders in traffic simulation systems improving transportation logistics.” (2023), [Online]. Available: <https://www.kingsresearch.com/blog/top-5-traffic-simulation-systems-improving-logistics> (visited on 04/29/2025).
- [19] P. group. “Why would you use a traffic simulation?” (), [Online]. Available: <https://www.ptvgroup.com/en/application-areas/traffic-simulation> (visited on 04/29/2025).
- [20] G. Laemmel. “Simulation of urban mobility - sumo.” (2025), [Online]. Available: https://www.eclipse.org/community/eclipse_newsletter/2017/august/article2.php (visited on 04/29/2025).
- [21] W. Peter and other. “Simulation of urban mobility.” (2025), [Online]. Available: <https://eclipse.dev/sumo/> (visited on 04/29/2025).
- [22] G. A. C. (DLR) and others. “Sumo at a glance.” (2025), [Online]. Available: https://sumo.dlr.de/docs/SUMO_at_a_Glance.html (visited on 04/29/2025).
- [23] G. A. C. (DLR) and others. “Scenarios.” (2025), [Online]. Available: <https://sumo.dlr.de/docs/Data/Scenarios.html> (visited on 04/29/2025).
- [24] K. Schrab, R. Protzmann, and I. Radusch, “A large-scale traffic scenario of berlin for evaluating smart mobility applications,” in *Smart Energy for Smart Transport*, E. G. Nathanail, N. Gavanis, and G. Adamos, Eds., Cham: Springer Nature Switzerland, 2023, pp. 276–287.
- [25] M. Gueriau and I. Dusparic, “Quantifying the impact of connected and autonomous vehicles on traffic efficiency and safety in mixed traffic,” in *23rd IEEE International Conference on Intelligent Transportation Systems*, 2020.
- [26] M. Rapelli, C. Casetti, and G. Gagliardi, “Vehicular traffic simulation in the city of turin from raw data,” *IEEE Transactions on Mobile Computing*, pp. 1–12, 2021. DOI: 10.1109/TMC.2021.3075985.

- [27] O. K. Tonguz. “How vehicle-to-vehicle communication could replace traffic lights and shorten commutes.” (2018), [Online]. Available: <https://spectrum.ieee.org/how-vehicle-to-vehicle-communication-could-replace-traffic-lights-and-shorten-commutes> (visited on 04/29/2025).
- [28] C. Bachechi, L. Po, and F. Rollo, “Big data analytics and visualization in traffic monitoring,” *Big Data Research*, vol. 27, p. 100 292, 2022, ISSN: 2214-5796. DOI: <https://doi.org/10.1016/j.bdr.2021.100292>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S221457962100109X>.
- [29] K. Daniel, E. Jakob, B. Michael, and B. Laura, “Recent development and applications of sumo – simulation of urban mobility,” *International Journal on Advances in Systems and Measurements*, vol. 5, pp. 128–138, 2012. [Online]. Available: https://sumo.dlr.de/pdf/sysmea_v5_n34_2012_4.pdf.
- [30] L. Da, C. Chu, W. Zhang, and H. Wei, *Cityflower: An efficient and realistic traffic simulator with embedded machine learning models*, 2024. arXiv: 2402.06127 [cs.MA]. [Online]. Available: <https://arxiv.org/abs/2402.06127>.
- [31] H. Zhang, S. Feng, C. Liu, *et al.*, “Cityflow: A multi-agent reinforcement learning environment for large scale city traffic scenario,” in *The World Wide Web Conference*, ser. WWW ’19, ACM, May 2019, pp. 3620–3624. DOI: 10.1145/3308558.3314139. [Online]. Available: <http://dx.doi.org/10.1145/3308558.3314139>.
- [32] K. Kušić, R. Schumann, and E. Ivanjko, “A digital twin in transportation: Real-time synergy of traffic data streams and simulation for virtualizing motorway dynamics,” *Advanced Engineering Informatics*, vol. 55, p. 101 858, 2023, ISSN: 1474-0346. DOI: <https://doi.org/10.1016/j.aei.2022.101858>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1474034622003160>.
- [33] M. Napolitano, A. Somma, A. De Benedictis, and N. Mazzocca, “Flowtwin: A digital twin for traffic flow monitoring,” *IEEE Open Journal of Intelligent Transportation Systems*, vol. 6, pp. 1551–1568, 2025. DOI: 10.1109/OJITS.2025.3637341.
- [34] J. Wen, J. Hu, C. Wu, X. Xiao, and N. Lyu, “A novel stochastic second-order macroscopic continuum traffic flow model for traffic instability,” *Chaos, Solitons & Fractals*, vol. 190, p. 115 752, 2025, ISSN: 0960-0779. DOI: <https://doi.org/10.1016/j.chaos.2024.115752>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0960077924013043>.
- [35] P. Burggräf, T. Adlon, R. Minderjahn, N. Schäfer, and T. Fassmer, “Application of stochastic risk simulation to increase depth of production planning,” *International Journal of Naval Architecture and Ocean Engineering*, vol. 15, p. 100 545, 2023, ISSN: 2092-6782. DOI: <https://doi.org/10.1016/j.ijnaoe.2023.100545>. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2092678223000341>.
- [36] J. Y. Yen, “Finding the k shortest loopless paths in a network,” *Management Science*, vol. 17, no. 11, pp. 712–716, 1971, ISSN: 0025-1909, 1526-5501. [Online]. Available: <http://www.jstor.org/stable/2629312> (visited on 04/16/2026).
- [37] J. de Dios Ortúzar and L. Willumsen, *Modelling Transport*. Wiley, 2011, ISBN: 9781119993520. [Online]. Available: <https://books.google.be/books?id=qWa5MyS4CiwC>.

- [38] Transportation Research Board, *Highway capacity manual*, National Academies of Sciences, Engineering, and Medicine, Washington, D.C., 2000.
- [39] A. Wilson, “A statistical theory of spatial distribution models,” *Transportation Research*, vol. 1, no. 3, pp. 253–269, 1967, ISSN: 0041-1647. DOI: [https://doi.org/10.1016/0041-1647\(67\)90035-4](https://doi.org/10.1016/0041-1647(67)90035-4). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/0041164767900354>.